

# Principal Component Analysis: Solved Examples Using Eigenvalues and Eigenvectors

## Introduction

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms correlated variables into a set of uncorrelated principal components using eigenvalues and eigenvectors of the covariance matrix.

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## Example 1: Basic PCA with 2D Data

### Problem

Given the following dataset with 2 variables ( $x_1$ ,  $x_2$ ) and 4 observations:

| Observation | $x_1$ | $x_2$ |
|-------------|-------|-------|
| 1           | 4     | 2     |
| 2           | 2     | 3     |
| 3           | 3     | 4     |
| 4           | 5     | 5     |

Perform PCA and find the principal components.

## Solution

### Step 1: Center the data (subtract mean from each variable)

Mean of  $x_1$ :  $\mu_1 = (4 + 2 + 3 + 5)/4 = 3.5$

Mean of  $x_2$ :  $\mu_2 = (2 + 3 + 4 + 5)/4 = 3.5$

Centered data:

- $(4 - 3.5, 2 - 3.5) = (0.5, -1.5)$
- $(2 - 3.5, 3 - 3.5) = (-1.5, -0.5)$
- $(3 - 3.5, 4 - 3.5) = (-0.5, 0.5)$
- $(5 - 3.5, 5 - 3.5) = (1.5, 1.5)$

### Step 2: Calculate the covariance matrix

Covariance matrix  $C = (1/n)X^T X$ , where  $n = 4$

$C = (1/4) \times$

$$\begin{bmatrix} 0.5 & -1.5 & -0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \\ -0.5 & 0.5 & & \\ 1.5 & 1.5 & & \end{bmatrix} \begin{bmatrix} 0.5 & -1.5 \\ -1.5 & -0.5 \\ -0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

After multiplication:

$$C = \begin{bmatrix} 1.25 & 1.25 \\ 1.25 & 1.75 \end{bmatrix}$$

### Step 3: Find eigenvalues

$$\text{Det}(C - \lambda I) = 0$$

$$|(1.25 - \lambda) \quad 1.25| = 0$$

$$|1.25 \quad (1.75 - \lambda)|$$

$$(1.25 - \lambda)(1.75 - \lambda) - 1.25^2 = 0$$

$$2.1875 - 1.25\lambda - 1.75\lambda + \lambda^2 - 1.5625 = 0$$

$$\lambda^2 - 3\lambda + 0.625 = 0$$

Using quadratic formula:

$$\lambda = (3 \pm \sqrt{(9 - 2.5)})/2 = (3 \pm \sqrt{6.5})/2$$

$\lambda_1 = 2.77$  (largest eigenvalue - 1st principal component)  $\lambda_2 = 0.23$  (smallest eigenvalue - 2nd principal component)

#### Step 4: Find eigenvectors

For  $\lambda_1 = 2.77$ :

$$(C - 2.77I)v = 0$$

$$\begin{bmatrix} -1.52 & 1.25 \end{bmatrix} \begin{bmatrix} v_1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{bmatrix} 1.25 & -1.02 \end{bmatrix} \begin{bmatrix} v_2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

From first equation:  $-1.52v_1 + 1.25v_2 = 0$

$$v_2 = 1.22v_1$$

Normalizing ( $v_1^2 + v_2^2 = 1$ ):

$$v_1^2 + (1.22v_1)^2 = 1$$

$$v_1 \approx 0.634, v_2 \approx 0.773$$

**First eigenvector:  $e_1 = [0.634, 0.773]$**

For  $\lambda_2 = 0.23$ : Following similar process: **Second eigenvector:  $e_2 = [-0.773, 0.634]$**

#### Step 5: Variance explained

$$\text{Total variance} = \lambda_1 + \lambda_2 = 2.77 + 0.23 = 3.0$$

Variance explained by PC1 =  $2.77/3.0 = 92.3\%$  Variance explained by PC2 =  $0.23/3.0 = 7.7\%$

**Conclusion:** The first principal component captures 92.3% of the variance, so we can reduce dimensionality from 2D to 1D with minimal information loss.

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## Example 2: 3D to 2D Reduction

### Problem

Given a centered data matrix X (already mean-centered) with covariance matrix:

$$C = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Find the principal components and reduce to 2D.

### Solution

#### Step 1: Find eigenvalues

$$\text{Det}(C - \lambda I) = 0$$

$$\begin{vmatrix} 3-\lambda & 2 & 1 \\ 2 & 4-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

Expanding determinant:

$$\begin{aligned} & (3-\lambda)[(4-\lambda)(2-\lambda) - 0] - 2[2(2-\lambda) - 0] + 1[0 - (4-\lambda)] \\ &= (3-\lambda)(8 - 6\lambda + \lambda^2) - 2(4 - 2\lambda) - 4 + \lambda \\ &= 24 - 18\lambda + 3\lambda^2 - 8\lambda + 6\lambda^2 - \lambda^3 - 8 + 4\lambda - 4 + \lambda \\ &= -\lambda^3 + 9\lambda^2 - 21\lambda + 12 = 0 \end{aligned}$$

Solving (by factoring or numerical methods):  $\lambda_1 = 6.0$  (largest)  $\lambda_2 = 2.5$   $\lambda_3 = 0.5$  (smallest)

### Step 2: Find eigenvectors

For  $\lambda_1 = 6.0$ :  $\mathbf{e}_1 \approx [0.408, 0.816, 0.408]$

For  $\lambda_2 = 2.5$ :  $\mathbf{e}_2 \approx [0.707, 0, -0.707]$

For  $\lambda_3 = 0.5$ :  $\mathbf{e}_3 \approx [-0.577, 0.577, 0.577]$

### Step 3: Variance analysis

Total variance =  $6.0 + 2.5 + 0.5 = 9.0$

PC1:  $6.0/9.0 = 66.7\%$  PC2:  $2.5/9.0 = 27.8\%$  PC3:  $0.5/9.0 = 5.5\%$

Cumulative variance (PC1 + PC2) = **94.5%**

### Step 4: Projection to 2D

To project original data onto first two principal components:

$\mathbf{P} = [\mathbf{e}_1 \mid \mathbf{e}_2] = \begin{bmatrix} 0.408 & 0.707 \\ 0.816 & 0 \\ 0.408 & -0.707 \end{bmatrix}$

New coordinates:  $\mathbf{Y} = \mathbf{X} \times \mathbf{P}$

**Conclusion:** By keeping only the first two principal components, we retain 94.5% of the variance while reducing from 3D to 2D.

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## Key Formulas

1. **Covariance Matrix:**  $\mathbf{C} = (1/n)\mathbf{X}^T \mathbf{X}$  (for centered data)

2. **Eigenvalue Equation:**  $Cv = \lambda v$
  3. **Characteristic Equation:**  $\text{Det}(C - \lambda I) = 0$
  4. **Variance Explained:**  $\text{Variance}(\text{PC}_i) = \lambda_i / \sum \lambda_j$
  5. **Projection:**  $Y = X \times P$ , where  $P$  is the matrix of eigenvectors
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### Example 3: Student Performance Data

#### Problem

Three students are evaluated on two tests. The data matrix is:

| Student | Test 1 | Test 2 |
|---------|--------|--------|
| A       | 6      | 4      |
| B       | 8      | 6      |
| C       | 4      | 2      |

Perform PCA and find the principal components.

#### Solution

##### Step 1: Center the data

Mean of Test 1:  $\mu_1 = (6 + 8 + 4)/3 = 6$

Mean of Test 2:  $\mu_2 = (4 + 6 + 2)/3 = 4$

Centered data matrix  $X$ :

$$X = \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}$$

### Step 2: Covariance matrix

$$C = (1/n)X^T X = (1/3) \times$$

$$\begin{bmatrix} 0 & 2 & -2 \\ 0 & 2 & -2 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}$$

$$C = (1/3) \times \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} = \begin{bmatrix} 8/3 & 8/3 \\ 8/3 & 8/3 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} 8/3 & 8/3 \\ 8/3 & 8/3 \end{bmatrix}$$

$$C = \begin{bmatrix} 2.667 & 2.667 \\ 2.667 & 2.667 \end{bmatrix}$$

$$\begin{bmatrix} 2.667 & 2.667 \\ 2.667 & 2.667 \end{bmatrix}$$

### Step 3: Eigenvalues

$$\text{Det}(C - \lambda I) = 0$$

$$\begin{vmatrix} (2.667 - \lambda) & 2.667 \\ 2.667 & (2.667 - \lambda) \end{vmatrix} = 0$$

$$\begin{vmatrix} 2.667 & 2.667 \\ 2.667 & 2.667 \end{vmatrix} = 0$$

$$(2.667 - \lambda)^2 - (2.667)^2 = 0$$

$$\lambda^2 - 5.334\lambda = 0$$

$$\lambda(\lambda - 5.334) = 0$$

$$\lambda_1 = 5.334 \quad \lambda_2 = 0$$

### Step 4: Eigenvectors

For  $\lambda_1 = 5.334$ :

$$(C - 5.334I)v = 0$$

$$\begin{bmatrix} -2.667 & 2.667 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{bmatrix} 2.667 & -2.667 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

From first equation:  $-2.667v_1 + 2.667v_2 = 0$

$$v_1 = v_2$$

Normalizing:  $e_1 = [1/\sqrt{2}, 1/\sqrt{2}] = [0.707, 0.707]$

For  $\lambda_2 = 0$ :  $e_2 = [1/\sqrt{2}, -1/\sqrt{2}] = [0.707, -0.707]$

### Step 5: Interpretation

Variance explained by PC1 =  $5.334/5.334 = 100\%$  Variance explained by PC2 =  $0/5.334 = 0\%$

**Conclusion:** The data lies perfectly on a 1D line in 2D space. All students' performances are perfectly correlated across both tests. We can reduce to 1D with zero information loss!

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## Example 4: Feature Selection Problem

### Problem

Given the covariance matrix for three features:

$$C = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Determine how many principal components to retain to capture at least 90% of variance.



## Solution

### Step 1: Find eigenvalues

$$\text{Det}(C - \lambda I) = 0$$

$$\begin{vmatrix} 4-\lambda & 1 & 0 \\ 1 & 3-\lambda & 1 \\ 0 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)[(3-\lambda)(2-\lambda) - 1] - 1[1(2-\lambda) - 0] = 0$$

$$(4-\lambda)(6 - 5\lambda + \lambda^2 - 1) - (2 - \lambda) = 0$$

$$(4-\lambda)(\lambda^2 - 5\lambda + 5) - 2 + \lambda = 0$$

$$4\lambda^2 - 20\lambda + 20 - \lambda^3 + 5\lambda^2 - 5\lambda - 2 + \lambda = 0$$

$$-\lambda^3 + 9\lambda^2 - 24\lambda + 18 = 0$$

Solving numerically:  $\lambda_1 \approx 4.73$   $\lambda_2 \approx 3.00$   $\lambda_3 \approx 1.27$

### Step 2: Calculate variance explained

$$\text{Total variance} = 4.73 + 3.00 + 1.27 = 9.00$$

| PC  | Eigenvalue | Individual % | Cumulative % |
|-----|------------|--------------|--------------|
| PC1 | 4.73       | 52.6%        | 52.6%        |
| PC2 | 3.00       | 33.3%        | 85.9%        |
| PC3 | 1.27       | 14.1%        | 100.0%       |

**Answer:** We need **all three components** to reach 90% variance. PC1 and PC2 together only capture 85.9%.

If we're willing to accept 85.9% variance, we could use just 2 components.

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## Example 5: Correlation vs Covariance Matrix

### Problem

Given data with different scales:

| Observation | Height (cm) | Weight (kg) |
|-------------|-------------|-------------|
| 1           | 150         | 50          |
| 2           | 160         | 60          |
| 3           | 170         | 70          |
| 4           | 180         | 80          |

Compare PCA using covariance vs correlation matrix.

### Solution

#### Part A: Using Covariance Matrix

Step 1: Center data

Mean height = 165 cm, Mean weight = 65 kg

Centered:

[-15, -15]  
[-5, -5]  
[ 5, 5]  
[15, 15]

Step 2: Covariance matrix

$C = \begin{bmatrix} 166.67 & 166.67 \\ 166.67 & 166.67 \end{bmatrix}$

Eigenvalues:  $\lambda_1 = 333.33$ ,  $\lambda_2 = 0$

### Part B: Using Correlation Matrix

Step 1: Standardize data (subtract mean, divide by std)

Std(height) = 12.91, Std(weight) = 12.91

Standardized data is identical to centered data divided by std.

Step 2: Correlation matrix

$R = \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix}$

Eigenvalues:  $\lambda_1 = 2.0$ ,  $\lambda_2 = 0$

### Key Insight:

- With covariance matrix: Heights and weights have same scale, so results are similar
  - With correlation matrix: Variables are standardized first
  - When variables have different scales (e.g., height in mm vs weight in kg), use **correlation matrix** to avoid bias toward high-variance features
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## Example 6: Practical Dimensionality Reduction

### Problem

A dataset has 5 features with the following eigenvalues from PCA:

$$\lambda_1 = 125, \lambda_2 = 85, \lambda_3 = 40, \lambda_4 = 10, \lambda_5 = 5$$

How many components should we retain using:

- a) Kaiser criterion (eigenvalue > 1)
- b) 95% variance threshold
- c) Scree plot elbow method

### Solution

$$\text{Total variance} = 125 + 85 + 40 + 10 + 5 = 265$$

#### Cumulative Variance Table:

| PC  | Eigenvalue | Individual % | Cumulative % |
|-----|------------|--------------|--------------|
| PC1 | 125        | 47.2%        | 47.2%        |
| PC2 | 85         | 32.1%        | 79.2%        |
| PC3 | 40         | 15.1%        | 94.3%        |
| PC4 | 10         | 3.8%         | 98.1%        |
| PC5 | 5          | 1.9%         | 100.0%       |

**a) Kaiser Criterion:** Since we're using covariance matrix and eigenvalues aren't standardized, this isn't applicable. (Kaiser works with correlation matrix where  $\lambda > 1$  is meaningful)

**b) 95% Variance Threshold:** Retain **4 components** ( $98.1\% > 95\%$ )

**c) Scree Plot (Elbow Method):** Looking at eigenvalue drops:

- 125  $\rightarrow$  85: drop of 40
- 85  $\rightarrow$  40: drop of 45 (steep)
- 40  $\rightarrow$  10: drop of 30 (steep)
- 10  $\rightarrow$  5: drop of 5 (leveling off)

Elbow appears after PC3, suggesting retain **3 components** (94.3% variance).

**Recommendation:** Use 3 components for good balance between dimensionality reduction and information retention.

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## Example 7: Geometric Interpretation

### Problem

Data points: (2,0), (0,2), (-2,0), (0,-2)

Find principal components and interpret geometrically.

### Solution

**Step 1: Center data** (already centered at origin)

**Step 2: Covariance matrix**

$$C = (1/4) \times$$

$$\begin{bmatrix} 2 & 0 & -2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 \end{bmatrix}$$

$$C = (1/4) \times \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

### Step 3: Eigenvalues

$$\text{Det}(C - \lambda I) = (2-\lambda)(2-\lambda) - 0 = 0$$

$$(2-\lambda)^2 = 0$$

$$\lambda_1 = \lambda_2 = 2 \text{ (equal eigenvalues!)}$$

### Step 4: Eigenvectors

Any two orthogonal unit vectors work:  $\mathbf{e}_1 = [1, 0]$  (x-axis direction)  $\mathbf{e}_2 = [0, 1]$  (y-axis direction)

**Geometric Interpretation:** The data forms a perfect circle/square pattern. There's no preferred direction of variance - all directions are equally important! The data has **rotational symmetry**.

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## Practice Problems (Try These Yourself!)

### Problem A

Covariance matrix:  $C = \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$

Find eigenvalues and eigenvectors.

**Hint:**  $\text{Det}(C - \lambda I) = \lambda^2 - 8\lambda + 11 = 0$

### **Problem B**

Eigenvalues are 100, 25, 10, 5, 2. What percentage of variance is captured by the first two components?

### **Problem C**

If PCA on 10 features yields eigenvalues: 50, 30, 15, 10, 8, 5, 3, 2, 1, 1, how many components would you retain and why?

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## **Summary**

PCA transforms data into a new coordinate system where:

- Principal components are orthogonal (uncorrelated)
- Components are ordered by variance explained
- First few components capture most variation
- Dimensionality reduction with minimal information loss

The eigenvalues represent the variance along each principal component, and eigenvectors define the direction of maximum variance.

### **Key Decision Rules:**

- Retain components explaining 90-95% cumulative variance
- Use scree plot elbow method
- For correlation matrix: Kaiser criterion ( $\lambda > 1$ )
- Consider interpretability and downstream task requirements